Strategic Assessment of Foundation Models for Moderating Indonesian Disinformation and Sociocultural Harm

The digital landscape of Indonesia, characterized by over 210 million internet users and a complex tapestry of more than 700 local languages, presents a formidable challenge for automated content moderation. The rapid dissemination of disinformation, *fitnah* (slander), and *kebencian* (hate speech) requires more than simple keyword filtering; it demands a deep linguistic understanding and the ability to reason through sociocultural nuances.[1] Large Language Models (LLMs) have evolved into the primary technology for addressing these challenges, offering the capability for continual pre-training (CPT) and supervised fine-tuning (SFT) to align global knowledge with local sensitivities.[3] This report evaluates three premier open-source base models—Qwen3, the Sahabat-AI ecosystem (based on Llama 3.1), and SeaLLMs-v3—assessing their architectural suitability and empirical validation for the Indonesian context.

The Sociopolitical and Linguistic Imperative in Indonesia

In Indonesia, the spread of harmful content is not merely a technical issue but a social one that can trigger real-world conflicts. Disinformation often manifests as "provocative hoaxes," designed to incite anger or panic.[5] The term *fitnah* specifically refers to malicious slander that targets individuals or groups, often amplified through private messaging apps and social media.[7] Furthermore, the geographic distribution of vulnerability varies; urban populations are frequently targets of sophisticated multimodal manipulation (deepfakes and visual hoaxes), while rural areas remain susceptible to textual disinformation.[1]

Automating the detection of these phenomena requires models that can handle the "agglutinative" nature of Bahasa Indonesia—where prefixes and suffixes drastically alter root meanings—and the prevalent use of colloquial dialects.[4] The failure of Western-centric models to identify Indonesian "slang" or regional idioms often leads to the generation of unsafe outputs or the missing of toxic content.[4] Therefore, the selection of a base model must prioritize those with high tokenization efficiency and a proven track record in Southeast Asian (SEA) benchmarks.[9]

Key Terminology	Context in Indonesia	Technical Requirement

Disinformation	Politically motivated hoaxes and false narratives.[7]	High factual accuracy and world knowledge.[12]
Fitnah (Slander)	Malicious claims intended to ruin reputation.[6]	Logical reasoning to identify factual inconsistencies.[13]
Kebencian (Hate Speech)	Attacks on SARA (Ethnicity, Religion, Race, Inter-group).[8]	Nuanced toxicity detection across dialects.[4]

Candidate I: The Qwen3 Series (Reasoning and Multilingual Scale)

The Qwen series, developed by Alibaba Cloud, has established itself as a leader in reasoning-intensive tasks. The latest iteration, Qwen3, represents a significant advancement in the density of pre-training and architectural efficiency.[13] For a project focused on classification and reasoning of complex disinformation, the Qwen3 series provides a robust foundation due to its "Thinking Mode" and massive multilingual corpus.[12]

Architectural Innovations and Pre-training Scale

Qwen3 models are trained on an unprecedented 36 trillion tokens, covering 119 languages and dialects, including a substantial representation of the Austronesian family (Indonesian, Javanese, Sundanese, and Balinese).[13] This pre-training scale is nearly double that of previous generations, ensuring a deeper reservoir of factual knowledge necessary for identifying disinformation.[12] The architecture implements Grouped Query Attention (GQA) and SwiGLU activation functions, which optimize inference throughput and training stability.[12]

A defining feature of the Qwen3 series is its dual-mode capability: "Thinking Mode" and "Non-Thinking Mode".[13] In the Thinking Mode, the model utilizes step-by-step reasoning to solve complex problems, a feature that is critical for identifying the logical fallacies often present in *fitnah* or slanderous narratives.[13] The model can essentially "reason" through a claim before providing a classification, mirroring the human fact-checking process.[13]

Empirical Performance and Validation

Validation of Qwen's capabilities comes from several high-level benchmarks. On the GPQA Diamond benchmark—a test of doctoral-level science reasoning—Qwen3-235B-A22B achieved a score of 88.4, the highest among open-source models as of 2026.[17] This suggests an advanced capacity for processing complex information that exceeds general conversational ability.[17] In the Artificial Analysis Multilingual Index, Qwen models consistently rank alongside

top proprietary models for Indonesian language reasoning.[19]

Benchmark	Qwen3-235B-A22B	Qwen3-8B	Llama 3.1-8B
MMLU-Pro	84.4 [17]	~78.0 [18]	73.0 (Est.) [15]
GPQA Diamond	81.1 [17]	78.1 [18]	69.8 [18]
AIME 2025	92.3 [17]	82.7 [18]	TBD [18]
Chatbot Arena	1422 [17]	1365 [18]	1328 [18]

The Qwen3-235B-A22B uses a Mixture-of-Experts (MoE) design with 235 billion total parameters, of which only 22 billion are active per token.[15] This allows it to maintain the reasoning power of a massive model while remaining computationally viable for high-volume classification tasks.[15] For Indonesian tasks, Qwen 2 and 3 have been noted for their fluency and tone sensitivity, slightly outperforming competitors in general knowledge and instruction following.[21]

Justification for Disinformation and Fitnah Analysis

The core justification for Qwen3 lies in its "Thinking Deeper, Act Faster" paradigm.[13] Disinformation often relies on subtle distortions that require a model to cross-reference multiple facts within its weights.[12] Qwen's strength in technically intensive areas—scoring 8.7 in coding and structured logic compared to Llama's 7.5—indicates a superior ability to follow complex rules for classification.[21] In the context of "fitnah," where the model must determine if a statement follows a pattern of malicious intent or factual falsehood, Qwen’s reasoning depth provides the necessary transparency.[21]

Candidate II: The Sahabat-AI Ecosystem (Localization and Safety)

While global models provide scale, the Sahabat-AI project represents the pinnacle of Indonesian localization. Sahabat-AI is an open-source ecosystem co-initiated by GoTo Group, Indosat Ooredoo Hutchison, and AI Singapore, specifically designed to empower the Indonesian AI landscape.[23] By taking the Meta Llama 3.1 architecture and applying rigorous CPT and SFT, the Sahabat-AI team has created a model series that is uniquely attuned to the sociocultural fabric of the archipelago.[25]

Strategic Localization and Dialect Support

Sahabat-AI models, such as the Llama-Sahabat-AI-v2-70B-IT and Gemma2-9B-CPT-Sahabat-AI, are fine-tuned on hundreds of thousands of Indonesian-specific instruction pairs.[25] Unlike general multilingual models, Sahabat-AI includes dedicated training pools for Javanese, Sundanese, Batak Toba, and Balinese.[25] This is critical for detecting *kebencian* (hate speech), which often moves from formal Bahasa Indonesia to colloquial or regional languages to evade standard filters.[4]

The training data includes approximately 448,000 Indonesian instruction-completion pairs, along with nearly 200,000 pairs in Javanese and Sundanese.[26] This depth of localization allows the model to understand "unwritten social norms" and "diverse local traditions," which are essential for distinguishing between cultural expression and harmful slander.[4]

Empirical Performance and Safety Validation

The Sahabat-AI models are validated using the SEA-HELM (Southeast Asian Holistic Evaluation of Language Models) and IndoMMLU benchmarks.[25] SEA-HELM evaluates models across six competencies: NLU, NLG, reasoning, linguistic diagnostics, instruction following, and multi-turn chat.[24] Critically, it includes a "Safety" pillar that specifically targets toxicity detection in Indonesian.[14]

Model Variant	Overall (ID+JV+SU)	Bahasa Indonesia	Javanese	Sundanese
Sahabat-AI-v1-9B	61.169 [26]	64.154 [26]	~56.00 [27]	~58.00 [27]
Gemma-2-9B (Base)	56.359 [26]	63.394 [26]	Lower	Lower
Llama-3.1-8B (Base)	45.866 [26]	51.944 [26]	Lower	Lower
SeaLLM-v2.1-8B	46.880 [26]	54.579 [26]	~45.00	~45.00

Human evaluation of these models has shown that the structurally enhanced Sahabat-AI model outperforms other approaches in "detoxification"—the task of transforming toxic text into neutral expressions while preserving the original meaning.[31] While base Llama 3 models might effectively retain meaning, they often leave "residual toxicity" in the output, whereas Sahabat-AI

variants demonstrate a superior ability to reduce toxicity to safe levels.[31]

Justification for Hate Speech and Cultural Reasoning

The Sahabat-AI ecosystem is justified primarily by its cultural grounding and safety alignment. The project collaborates with major Indonesian universities (Gadjah Mada University, University of Indonesia) and media groups (Kompas Gramedia, Tempo) to ensure the training data reflects the current Indonesian sociopolitical reality.[24] For a project moderating *kebencian* and *fitnah*, the model's ability to handle colloquialisms and local dialects—supported by the IndoSafety dataset—is a decisive advantage.[4] Furthermore, the model's open-weight nature allows for private deployment, ensuring that sensitive Indonesian data never leaves local infrastructure.[33]

Candidate III: SeaLLMs-v3 (Regional Specialization and Trustworthiness)

The SeaLLMs (Southeast Asian Large Language Models) family, developed by DAMO Academy, represents a third-party effort to optimize AI for the specific linguistic challenges of Southeast Asia.[11] SeaLLMs-v3 is the latest iteration, offering state-of-the-art performance in tasks like world knowledge, mathematical reasoning, and translation for 12 regional languages.[34]

Optimization for Southeast Asian Nuances

SeaLLMs-v3-7B is built upon the Qwen2-7B base but undergoes extensive language enhancement and specialized instruction tuning.[34] A key architectural advantage is its tokenization efficiency; SeaLLM can encode SEA languages with up to 9 times fewer tokens than standard models like GPT-4.[37] This efficiency directly translates to lower operational costs and faster inference times when processing large datasets of Indonesian social media content.[37]

The model is specifically designed to reduce hallucination—a common problem in disinformation detection—and to provide safe responses to queries related to SEA culture.[11] It utilizes a specially constructed instruction-tuning dataset that adapts to local laws and customs, ensuring that its "reasoning" about *fitnah* or *kebencian* is aligned with regional legal frameworks like Indonesia's UU ITE.[37]

Performance in Reasoning and World Knowledge

SeaLLMs-v3 excels in "Multilingual World Knowledge" benchmarks such as M3Exam, which consists of local exam questions from various SEA countries.[35] This benchmark is a proxy for how well the model understands the local history, geography, and social science context necessary to identify politically motivated hoaxes.[36]

Model	ID Accuracy (M3Exam)	Avg SEA (M3Exam)	ID MT-Bench (Avg)
SeaLLMs-v3-7B-Chat	0.558 [36]	0.592 [36]	6.66 [11]
Qwen2-7B-Instruct	0.558 [36]	0.579 [36]	5.89 [39]
SeaLLM-7B-v2.5	0.499 [36]	0.541 [36]	5.62 [39]
Gemma-7B (Base)	0.475 [36]	0.510 [36]	-

A critical validation point for SeaLLMs-v3 is its "Refusal-F1" score, which measures the model's ability to refuse answering questions about non-existent entities (a check for hallucination).[11] SeaLLMs-v3-7B-Chat achieved a Refusal-F1 of 68.95 for Indonesian, vastly outperforming models like Llama 3-8B-Instruct (3.91) and GLM-4-9B-Chat (1.97).[11] This high refusal rate is a strong indicator of the model's trustworthiness when evaluating potentially false claims in disinformation.[11]

Justification for Trustworthy Content Moderation

The justification for SeaLLMs-v3 rests on its specialized focus on trustworthiness and reduced hallucinations.[11] In the realm of disinformation, a model that generates "plausible-sounding" but false information is more dangerous than one that simply fails.[11] SeaLLMs-v3's high refusal score and its performance on SeaBench—a native-speaker-validated instruction set—prove its reliability for high-stakes content moderation.[11] Additionally, its ability to handle multiple SEA languages makes it suitable for monitoring transnational disinformation campaigns that target Indonesia from neighboring regions.[34]

Benchmarking Frameworks for Indonesian Validation

To prove that a chosen model is "good" for the project, it must be validated against benchmarks that reflect the Indonesian reality. The following frameworks are standard for professional-grade Indonesian NLP research.

IndoMMLU and IndoCareer

IndoMMLU is the primary benchmark for assessing a model's understanding of the Indonesian education system, covering 64 tasks from primary to university entrance levels.[23] It specifically tests for knowledge of local languages and cultures, areas where models like GPT-3.5 have

been shown to fail at the high school level.[30] IndoCareer expands this to professional and vocational expertise in Indonesia, covering fields like law and healthcare.[40] Models that score highly on these benchmarks demonstrate the "world knowledge" required to verify factual claims in news and social media.[30]

SEA-HELM (BHASA) and SeaExam

SEA-HELM (also known as BHASA) is a holistic suite developed by AI Singapore.[10] It focuses on culture, linguistics, safety, and multi-turn chat.[29] The SeaExam and SeaBench frameworks are distinct because they are not translated from English; they are constructed from real regional exams and native-speaker interactions.[38] This ensures that a model's high score is a result of native linguistic competence rather than a byproduct of its English-centric pre-training.[10]

Benchmark	Focus Area	Relevance to Project
IndoMMLU	Local education & culture.[30]	General knowledge for hoax verification.
SEA-HELM	Safety, Toxicity, Dialects.[14]	Detecting hate speech and regional "slang."
SeaExam	Native-constructed reasoning.[38]	Identifying cultural nuances in *fitnah*.
IndoSafety	Indonesian sociocultural safety.[4]	High-quality safety alignment for SFT.

Advanced Methodologies for CPT and SFT

Beyond selecting a base model, the project's success depends on the execution of the continual pre-training and supervised fine-tuning phases. Research suggests several advanced strategies to maximize performance in the Indonesian context.

Tokenization Efficiency: The TOBA LLM Framework

Indonesian morphology is highly productive, with root words combined with affixes to create new meanings.[9] The TOBA LLM research introduces a syllable-based tokenization framework that segments text at syllable boundaries.[9] This approach achieves a Rényi efficiency of 0.74, significantly higher than the 0.50-0.64 range of standard multilingual tokenizers.[9] By implementing a more efficient tokenizer during the CPT phase, the model can represent the

corpus with fewer tokens, shrinking the embedding layer and speeding up training convergence.[9]

Parallel Data and the OpenSeal Strategy

Building a regional LLM does not always require trillions of tokens of monolingual data. The OpenSeal project proved that adapting an English-only model (like OLMo 2) to Southeast Asian languages using only parallel data (34.7 billion tokens) can rival the performance of much larger multilingual models.[3] For the current project, utilizing parallel datasets of Indonesian and English news, or formal and colloquial Indonesian, can help the model learn the semantic relationships between different modes of expression used in disinformation.[2]

Contrastive Learning for Imbalanced Classification

Real-world disinformation data is often extremely imbalanced, with far fewer hoax samples than legitimate news.[7] Research into "Contrast-BERT" suggests that recasting classification as a contrastive learning task allows the model to learn representations unsupervisedly, mitigating bias.[7] This approach improved precision in Indonesian fake news classification by up to 26.64% while maintaining high recall.[7] Integrating contrastive objectives during the SFT phase could significantly improve the model's ability to catch rare but high-impact "fitnah".[7]

Explainable Hate Speech: The TARGE Framework

For content moderation, providing a binary label (Toxic/Clean) is often insufficient. The TARGE framework utilizes LLMs (specifically Mistral-7B) to generate explicit rationales for hate speech detection.[22] By identifying critical textual features and explaining why they are harmful, the model provides "faithful explainability" without compromising accuracy.[22] This is vital for transparency in moderation decisions, especially when dealing with the sensitive legal implications of *kebencian* in Indonesia.[8]

Implementation Strategy for Classification and Reasoning

The practical application of these models requires a two-stage approach: CPT for linguistic alignment and SFT for task-specific reasoning.

Phase I: Continual Pre-training for Local Context

The base model (Qwen, Sahabat-AI, or SeaLLM) should undergo CPT using a mixture of broad Indonesian corpora and specialized datasets. The OSCAR corpus (4 billion tokens) and CC-100 provide web-scale foundations, while the "Awesome Indonesian LLM Dataset" offers curated content spanning 15+ categories.[2]

During CPT, the focus should be on:

1. **Linguistic Diversity**: Incorporating the 10+ local languages beyond Bahasa Indonesia supported by the Awesome Indonesian repository.[2]
2. **Domain Expertise**: Including text from government fact-checking initiatives like Mafindo (Masyarakat Anti Fitnah Indonesia) to expose the model to the "implicit tone" of hoaxes.[7]
3. **Tokenization Alignment**: Optionally updating the tokenizer to better reflect Indonesian syllable structures to reduce inference latency.[9]

Phase II: Supervised Fine-tuning for Reasoning

The SFT phase should be guided by the reasoning requirements of the project. Using the IndoNLI (Natural Language Inference) dataset allows the model to learn complex reasoning by determining relationships between premises and hypotheses.[2] This is the foundation for detecting *fitnah*, as the model must verify if a claim follows logically from known facts.

For classification, SFT should include:

1. **Toxicity and Hate Speech**: Fine-tuning on NusaX (sentiment and toxicity) and IndoSafety to align the model with Indonesian sociocultural norms.[2]
2. **Multi-turn Reasoning**: Utilizing datasets like ID-MoCQA (Indonesian Multi-hop Question Answering) to probe if the model can reason through interrelated cultural concepts.[28]
3. **Detoxification Objectives**: Training the model to not only classify but also neutralize harmful content, contributing to a "more peaceful society".[31]

Comparative Analysis and Tactical Recommendations

The three models evaluated each serve a specific strategic role in the detection of disinformation and hate speech.

Selection Metric	Qwen3-235B-A22B	Sahabat-AI-v2 (70B)	SeaLLMs-v3-7B
Reasoning Power	Exceptional (S-Tier Reasoning).[17]	Strong (Enterprise RAG ready).[43]	High for its size (SOTA 7B).[11]
Dialect Depth	Broad Austronesian support.[13]	Deep local dialect fine-tuning.[25]	Focused on 12 SEA languages.[34]
Safety Alignment	General safety & refusal modes.[42]	Specific Indonesian safety (SEA-HELM).[14]	Enhanced trustworthy Refusal-F1.[11]

Operating Cost	High (Requires H100/A100 clusters).[42]	Medium (4x NVIDIA L40s).[25]	Low (Single RTX 4090 viable).[33]

The "Thinking" Approach to Disinformation

If the project prioritizes the absolute accuracy of reasoning for complex *fitnah* cases, Qwen3-235B-A22B is the primary recommendation.[15] Its superior scores on GPQA Diamond and MMLU-Pro indicate a level of intellectual depth that is unmatched in the open-source community.[17] The "Thinking Mode" acts as a built-in reasoning agent, allowing the model to decompose a slanderous claim into its constituent parts for verification.[13]

The "Cultural" Approach to Hate Speech

For the moderation of *kebencian* and localized slander, Sahabat-AI-v2-70B-IT is the most viable candidate.[25] Its development was explicitly geared toward the Indonesian social context, and it has undergone the most rigorous safety evaluation for the region.[14] Its ability to handle Batak Toba, Balinese, and other dialects ensures that harmful content cannot "hide" behind regionalisms.[25]

The "Efficient" Approach to Scale

For large-scale, real-time monitoring of social media feeds, SeaLLMs-v3-7B offers the best balance of speed and trust.[35] Its specialized pre-training for SEA languages and its state-of-the-art refusal rates (Refusal-F1) make it a highly reliable "first-line" classifier.[11] It is particularly strong at identifying hallucinations and ensuring that its factual claims about Indonesian culture are grounded in reality.[35]

Conclusion: Strategic Path Forward

The analysis of the current LLM landscape identifies a convergence of high-level reasoning and deep cultural localization. Qwen3 provides the raw reasoning capacity required to combat the most sophisticated forms of disinformation.[15] Sahabat-AI offers the necessary sociocultural grounding to navigate the sensitive landscape of Indonesian hate speech and regional dialects.[4] SeaLLMs-v3 provides a highly efficient and trustworthy mechanism for processing the vast quantities of data typical of the Indonesian digital sphere.[11]

To implement these findings, the project should focus on a hybrid architecture. A lightweight model like SeaLLMs-v3 can serve as a primary filter for toxicity and simple hoaxes, while more complex or ambiguous cases of *fitnah* can be routed to the larger reasoning engines of Qwen3 or Sahabat-AI. By utilizing regional benchmarks like IndoMMLU and SEA-HELM for continuous validation, and by incorporating advanced SFT techniques like contrastive learning and explainable rationales, the project can develop a world-class system for protecting the

Indonesian digital community from the harms of disinformation and hate speech.[7] The availability of these powerful open-source models ensures that Indonesia can maintain data sovereignty while deploying frontier-level AI capabilities.[33]

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